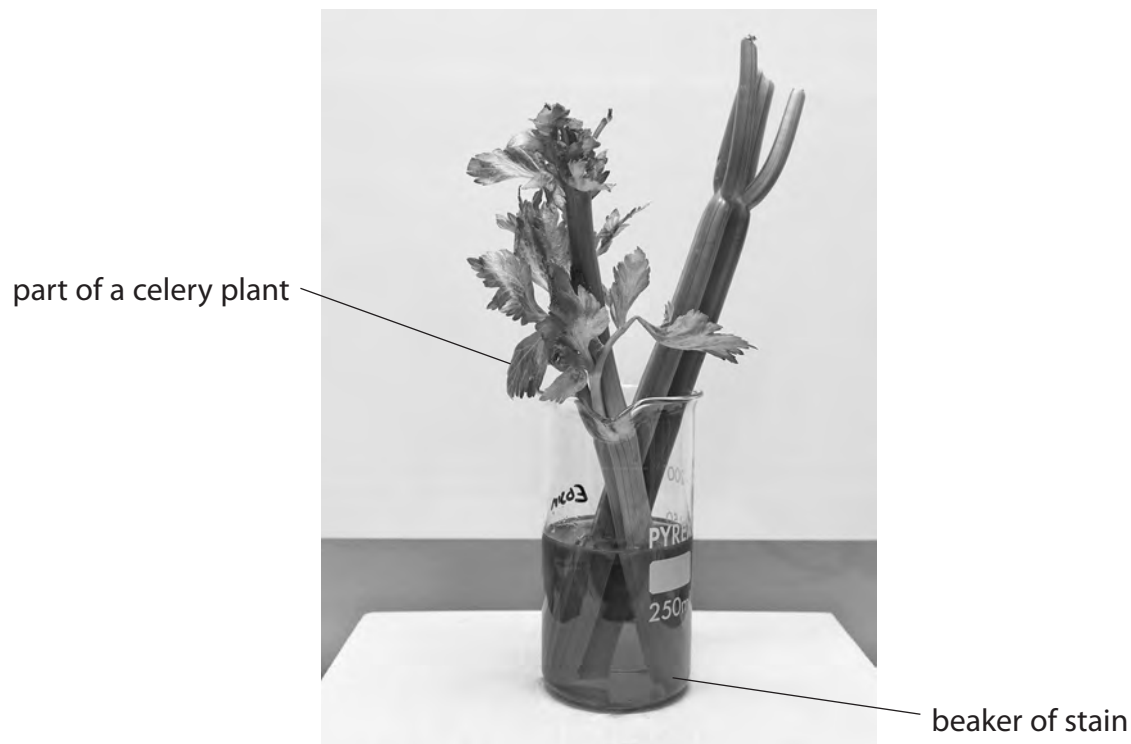


Answer ALL questions.

Write your answers in the space provided.

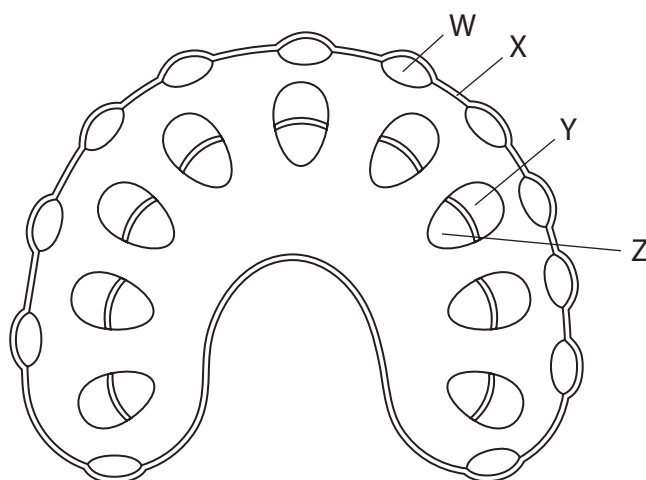
Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 The photograph shows part of a celery plant which has been placed into a beaker of stain.



(Source: © Sara Sadler/Alamy Stock Photo)

A student made a drawing of a cross-section of the bottom of the stem of the celery plant before it was placed into the beaker of stain.



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) One hour after being placed into the beaker of stain, the student observed a cross-section from half way up the celery stem.

(i) Which of the following would be a different colour due to the presence of the stain?

(1)

☐ **A** W

☐ **B** X

☐ **C** Y

☐ **D** Z

(ii) Explain why, after one hour, this part of the stem would be a different colour.

(2)

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(b) Cells in the leaf contain organelles that store molecules of starch.

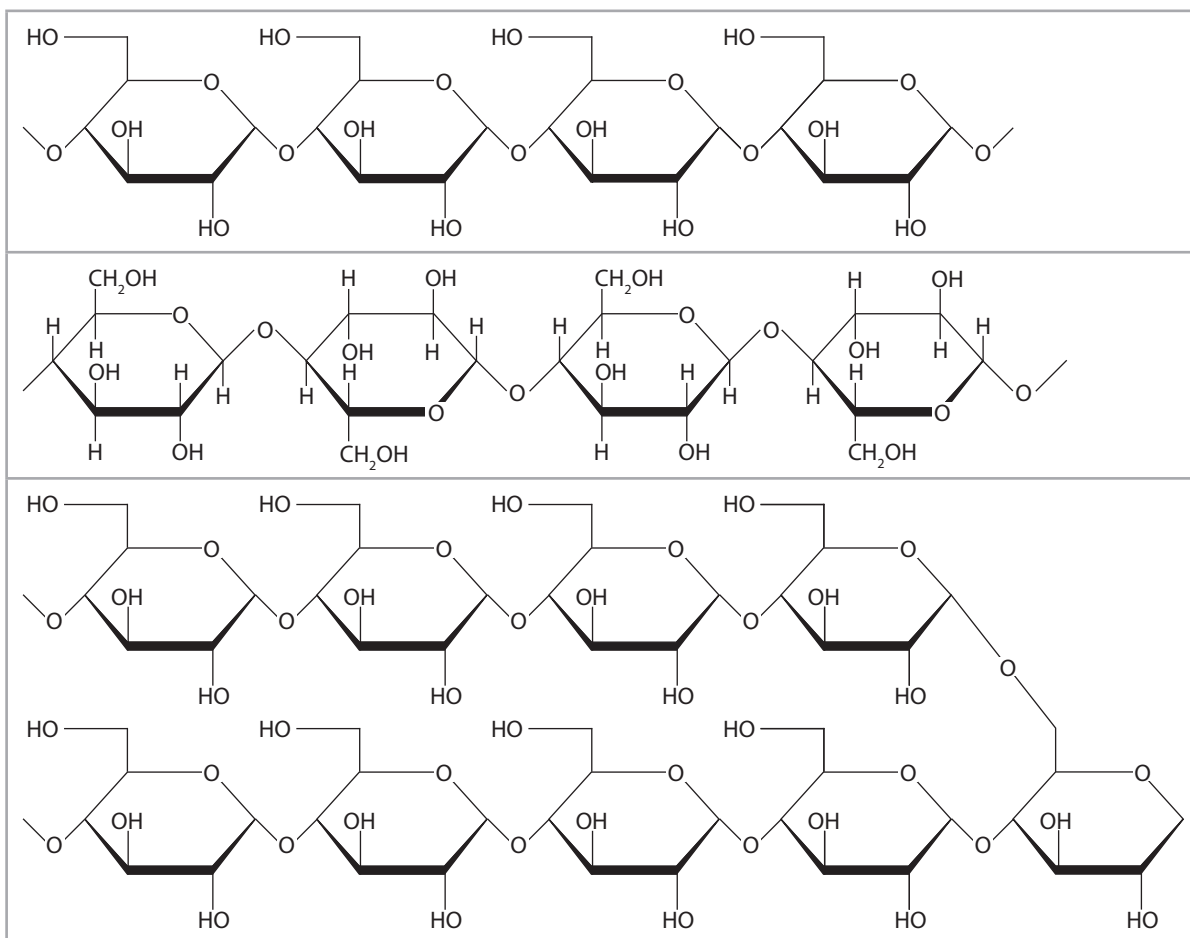
(i) Name **one** organelle in a leaf cell that stores starch.

(1)

.....



(ii) How many of the following show the structure of part of a starch molecule?



(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(iii) Give **one** role of starch molecules in a leaf cell.

(1)

.....

.....

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(c) Transparent plastic film can be made from oil-based plastics or from starch.

The photograph shows transparent bio-plastic film, made from starch.



transparent
bio-plastic film

(Source: © Roman/Alamy Stock Photo)

Give **one** reason why the use of starch-based transparent bio-plastic film is more sustainable than the use of oil-based plastics.

(1)

(Total for Question 1 = 7 marks)

